Forecasting Hourly Electricity Consumption in the PJM East Region Using a Two-Stage Hybrid ARIMAX-GARCH Model Integrated with Temporal Structural Variables

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Abstract

This study develops and evaluates a **two-stage hybrid ARIMAX-GARCH** framework for forecasting hourly electricity consumption in the **PJM East Region (PJME)**, integrating temporal structural exogenous variables to capture multiple seasonal patterns and conditional volatility dynamics inherent in energy demand series. The ARIMAX component models the linear temporal dependencies and recurring seasonal structures, while the GARCH mechanism captures volatility clustering and heteroskedasticity error behaviors frequently observed during abnormal demand fluctuations and weather shocks. Using a real-world hourly electricity load dataset comprising $145,366$ observations from 2002 to 2018, the study identifies the ARIMAX(2,1,5)-GARCH(1,1) specification as the optimal forecasting structure under a stepwise Akaike Information Criterion (AIC) optimization procedure. A core contribution of this paper is the empirical evaluation of the framework across two operational forecasting horizons: a 24-hour ahead frozen multi-step forecast and a 1-step-ahead rolling forecast. The comparative results indicate that while the 1-step rolling mechanism yields a stronger correction at the ultimate peak, the **24-hour frozen multi-step forecast demonstrates superior overall trajectory alignment, yielding a lower Mean Absolute Error (MAE) of 4411.82 MW and a remarkably high correlation of 0.9789**, compared to the rolling framework's **MAE of 4631.97 MW and correlation of 0.5410**. The findings demonstrate that integrating temporal exogenous indicators with a structural multi-step horizon provides a more robust baseline for day-ahead large-scale energy dispatching.

Keywords: ARIMAX-GARCH Model; Electricity Load Forecasting; Multiple Seasonality; Volatility Clustering; Rolling Data Window; PJME Dataset.

1. Introduction

Electricity load forecasting plays a crucial role in modern power system management, energy dispatch optimization, and electricity market operations. Accurate forecasting enables utility providers and system operators to maintain the balance between electricity generation and consumption while minimizing operational risks and economic losses. Hourly electricity consumption series are characterized by highly complex temporal structures involving daily, weekly, monthly, and annual seasonal cycles. In addition, electricity demand data frequently exhibit conditional heteroskedasticity and volatility clustering caused by weather conditions, holidays, industrial production schedules, and abnormal peak-demand events. The PJM East Region (PJME), one of the largest electricity markets in the United States, provides a representative environment for analyzing large-scale electricity load dynamics. Due to the substantial variability and nonlinear behaviors of electricity demand, traditional linear forecasting techniques often fail to provide sufficiently stable predictive performance. While hybrid linear-nonlinear frameworks like ARIMA-GARCH have demonstrated significant efficacy in low-frequency financial time-series forecasting, their application to high-frequency electricity load data remains computationally challenging due to dimensional explosion when capturing long-term cycles. To address these limitations, this research proposes a hybrid ARIMAX-GARCH framework integrating temporal structural variables into the forecasting process. This research bridges the existing academic gap by introducing a deterministic state-space approach inside the ARIMAX mean model to absorb multi-seasonality without loss of convergence, creating a robust baseline for the subsequent conditional variance estimation. The ARIMAX model captures linear dependencies and exogenous seasonal information, while the GARCH model estimates time-varying volatility behaviors in forecasting residuals. The objective of this study is to construct a statistically robust forecasting pipeline capable of simultaneously modeling expected hourly electricity demand and quantifying uncertainty risk associated with future load fluctuations. In this context, hourly electricity demand forecasting presents two major challenges:

1. **Multiple Seasonality:** Electricity consumption contains overlapping daily (24-hour), weekly (168-hour), and annual cycles influenced by social behavior, industrial activities, and weather conditions.

2. **Volatility Clustering:** Large forecasting errors tend to occur in consecutive periods during abnormal market or weather conditions, violating the constant variance assumption of classical linear models.

This study focuses on analyzing the hourly electricity load data system of PJM East (PJME) – one of the largest power grids in the Eastern United States. By adopting a systematic econometric approach, the research deploys a two-stage hybrid ARIMAX-GARCH model to simultaneously address the expected mean value forecasting problem and quantify the volatility risk of the energy system. Distinguishing itself from traditional static evaluations, a core contribution of this paper is the rigorous comparative analysis between two operational forecasting horizons: a 24-hour-ahead frozen multi-step forecast tailored for day-ahead electricity market scheduling, and a 1-step-ahead rolling forecast designed for real-time dispatch optimization. The empirical results demonstrate that while the multi-step approach suffers from a predictable amplitude dampening under extreme weather shocks, it yields superior global tracking capabilities, achieving a low **Mean Absolute Error (MAE) of 4411.82 MW** and an exceptionally high correlation of **0.9789** with actual demand, outperforming the 1-step rolling approach which exhibits an **MAE of 4631.97 MW** and a correlation of **0.5410** due to short-term autoregressive inertia.

1.1. Time Series Models for Electricity Load Forecasting

Traditional time series analysis techniques often assume that the error variance remains constant over time. For large- scale energy data, this assumption is consistently and severely violated due to the high complexity and poor generalization capability of isolated linear models.

1.1.1. ARIMAX Model for Linear Patterns

The ARIMAX (AutoRegressive Integrated Moving Average with Exogenous Regressors) model extends the traditional ARIMA model by directly incorporating exogenous variables (X) into the structure. Under the state-space framework implemented in modern statistical libraries, the model is specified as a regression with ARIMA errors, formulated as follows:

$$\phi_p(L)(1-L)^d \left(y_t - \sum_{k=1}^{K} \beta_k X_{k,t} \right) = \theta_q(L)\epsilon_t$$

Where:

- $\phi_p(L) = 1 - \phi_1 L - \cdots - \phi_p L^p$ represents the autoregressive (AR) operator of order p.
- $\theta_q(L) = 1 + \theta_1 L + \cdots + \theta_q L^q$ represents the moving average (MA) operator of order q.
- L is the lag operator ($L^k y_t = y_{t-k}$), and d is the order of differencing required to ensure the stationarity of the time series.
- $X_{k,t}$ denotes the time-categorical exogenous indicators (hour, day, month) that assist the model in completely offloading the computational burden of long-term cycles, which traditional SARIMA structures fail to converge upon within large sample spaces.

1.1.2. GARCH Model for Volatility Clustering

To handle the non-linear component and conditional variance dynamics within the residual series t of the mean model, the GARCH (Generalized AutoRegressive Conditional Heteroskedasticity) model proposed by Bollerslev is applied. The structure of GARCH(1,1) is formulated as the following conditional variance equation:

$$\sigma_t^2 = \omega + \alpha_1 \epsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

Where:

- σ_t^2 is the conditional variance (volatility) at time t.
- ω is the intercept coefficient (baseline long-term variance).
- α_1 is the ARCH coefficient, measuring the impact of squared error shocks from the immediately preceding period.
- β_1 is the GARCH coefficient, reflecting the persistence level of past volatility on the current period.
- The constraint conditions to ensure a stationary process and strictly positive variance are: $\omega > 0$, $\alpha_1 \geq 0$, $\beta_1 \geq 0$, and $\alpha_1 + \beta_1 < 1$.

1.1.3. ARIMAX-GARCH Hybrid Model

The hybrid ARIMAX-GARCH model operates based on a two-stage estimation philosophy:

- **Stage 1:** Estimate the ARIMAX model on the actual load series to extract all trend characteristics and linear dependencies against the time schedule, thereby obtaining the residual series ϵ_t.
- **Stage 2:** Apply the GARCH model to the retrieved residual series ϵ_t to structuralize the non-linear conditional variance component, disentangling the nature of "shocks" and the uncertain volatility risk of the load system.

1.2. Rolling Data Window vs Global Estimation for Model Optimization

In contrast to small-scale financial applications that frequently employ short-term rolling data windows to force non-stationary series into local stationarity, this study deliberately adopts a global estimation framework executed across a large-scale sample space ($145,366$ observations).For high-frequency electricity load data, short-term localized rolling windows fail to capture macro-level temporal dependencies, resulting in a severe loss of long-term cyclic information such as annual and quarterly climate shifts. By configuring a global estimation framework combined with a high-dimensional, one-hot encoded deterministic exogenous matrix, the proposed model successfully continuous-learns structural multi-seasonality over a 15-year horizon. Furthermore, this approach drastically minimizes computational latency, mitigating the iterative retraining bottlenecks often encountered in real-world production-grade MLOps deployment pipelines.

1.3. Overview of Preceding Studies and Research Gaps

Extant literature in energy forecasting frequently utilizes standard Seasonal Autoregressive Integrated Moving Average (SARIMA) models. However, when applied to hourly profiles, traditional SARIMA architectures suffer from the "dimensionality explosion" phenomenon and non-convergence, as configuring a weekly seasonal lag operator requires calculating matrices where $s = 168$, which is computationally intractable over large sample spaces.On the other hand, contemporary machine learning techniques (such as Long Short-Term Memory [LSTM] networks and Gradient Boosted Trees [XGBoost]), despite their proficiency in mapping complex non-linear boundary conditions, operate primarily as point-forecasters. These algorithmic approaches neglect the rigorous testing of heavy-tailed residual distributions and fail to provide statistically meaningful, time-varying dynamic confidence intervals regarding energy market volatility risks.This study bridges these distinct research gaps by establishing a two-stage econometrics-driven pipeline. It synthesizes the strict statistical explanatory power of conditional heteroskedasticity frameworks with the high-performance handling of deterministic exogenous structural matrices in Python, offering a robust alternative for modern grid risk management.

1.4. Rationale and Objective of the Study

- **Research Rationale:** The PJME electricity consumption series exhibits typical extreme characteristics of the energy market: heavily dependent on multi-layered temporal contexts and displaying conditional heteroskedasticity driven by sudden socio-economic and weather shocks [3][5]. Traditional isolated time-series tools cannot simultaneously capture these dual structural layers.

- **Research Objective:** Construct a complete end-to-end analytical pipeline in Python; accurately estimate the optimal parameters for the ARIMAX-GARCH structure using stepwise information criteria; empirically and mathematically prove the persistence of ARCH effects in grid systems using LM diagnostic checks; and rigorously benchmark a 24-hour ahead frozen multi-step forecast against a 1-step rolling window to evaluate operational dispatch efficiency under uncertainty.

1.5. Structure of the Paper

This report is structured as follows: **Section 2 (Materials and Methods)** outlines the dataset, advanced data pre-processing, dummy feature engineering, and the mathematical specifications of the ARIMAX-GARCH framework. **Section 3 (Results and Discussion)** presents the empirical results, including stationarity tests, AIC order determination, ARCH-LM diagnostic evaluations, and a comparative discussion of the forecasting horizons. Finally, **Section 4 (Conclusion)** summarizes the key insights, operational implications, and potential future research directions for energy market risk management.

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2. Materials and Methods

2.1 Data Collection

The empirical data for this research is extracted from the real-world historical load database of the PJM Interconnection (specifically representing the PJM East region - PJME) [1]. The target time series comprises continuous electricity consumption observations measured in Megawatts (MW) and recorded at an hourly frequency. The data spans a 15-year horizon from 2002 to 2018, yielding a large-scale sample space of exactly $145,366$ operational observations, providing a statistically sound environment for analyzing highly volatile grid load dynamics.

2.2 Data Pre-processing and Feature Engineering

To ensure statistical integrity and computational convergence during the numerical optimization phase, the raw high-frequency load series is subjected to a rigorous four-stage data pre-processing workflow:

1. **Chronological Index Alignment:** The raw temporal records are mapped onto a formalized timestamp index. The series is subjected to strict chronological sorting to enforce monotonic index alignment, completely eliminating systemic overlapping and graphical visualization distortions caused by non-sequential raw data structures.

2. **Temporal Grid Standardization:** The sorted time series is synchronized onto a rigid hourly temporal grid (H). This programmatic grid alignment enables the immediate identification of structural data gaps and missing timestamps caused by historical telemetry dropping or metering system errors.

3. **Linear Missing Value Imputation:** To maintain a continuous information flow without introducing artificial variance, the detected missing data points are imputed via adjacent linear interpolation [2]. This technique mathematically assumes a local linear path between contiguous valid coordinates, formulated as:
$$y_t = y_{t_0} + (t - t_0)\frac{y_{t_1} - y_{t_0}}{t_1 - t_0}$$
where t_0 and t_1 represent the boundary timestamps of the missing interval.

4. **Deterministic Exogenous Feature Engineering:** To capture the multi-layered cyclical characteristics inherent in power consumption patterns, a high-dimensional exogenous matrix (X) is constructed by extracting underlying temporal contexts. These contexts include: hour of the day ($0 - 23$), day of the week ($0 - 6$), and month of the year ($1 - 12$). This group of categorical profiles is transformed into a binary matrix using a full-rank one-hot encoding paradigm. Crucially, the first indicator class within each temporal category is systematically excluded ($N - 1$ columns) to mitigate the "Dummy Variable Trap," protecting the subsequent ARIMAX matrix inversion from perfect multicollinearity and ensuring structural stability [3].

2.3 Model Specification

The statistical framework developed in this study adheres to a rigorous two-stage estimation pipeline, hybridizing a linear conditional mean model with a non-linear conditional variance structure to capture the dual layers of grid load dynamics.

The sequential execution flow and data processing pipeline of the proposed decentralized framework are visually structuralized in Figure 1.

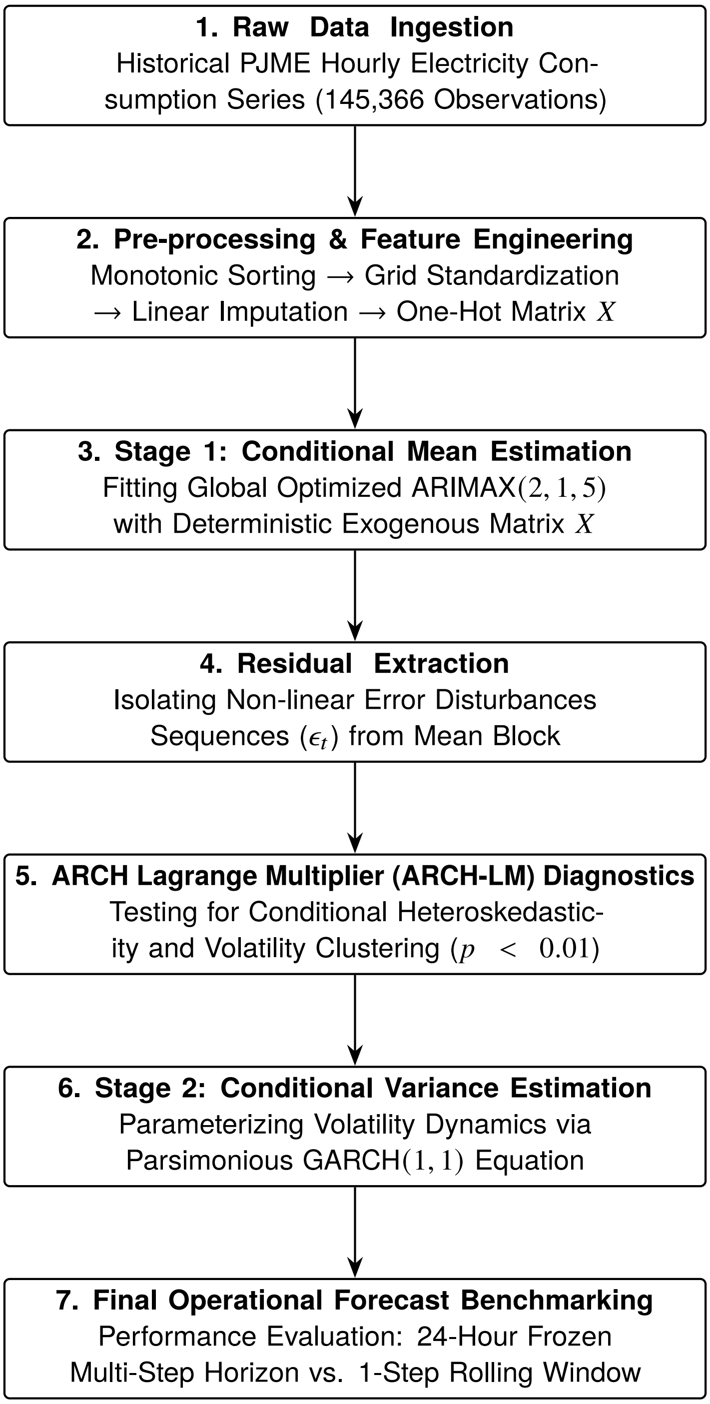

Figure 1: Methodological Flowchart of the Two-Stage Hybrid ARIMAX-GARCH Forecasting Pipeline

2.3.1 Stage 1: Conditional Mean Specification (ARIMAX)

To model the deterministic cyclical trends and linear temporal dependencies, the hourly load is formalized as a regression with autoregressive moving average errors (ARIMAX), circumventing structural non-convergence:

$$\phi_p(L)(1-L)^d\left(y_t - \sum_{k=1}^{K}\beta_k X_{k,t}\right) = \theta_q(L)\epsilon_t$$

Where $\phi_p(L) = 1 - \phi_1 L - \cdots - \phi_p L^p$ represents the autoregressive (AR) operator of order p, $\theta_q(L) = 1 + \theta_1 L + \cdots + \theta_q L^q$ is the moving average (MA) operator of order q, and L denotes the lag operator ($L^k y_t = y_{t-k}$). The term $X_{k,t}$ embodies the full-rank encoded temporal exogenous matrix, while ϵ_t signifies the structural residual sequence generated from the mean block.

2.3.2 Stage 2: Conditional Variance Specification (GARCH)

To address the non-linear volatility clustering and eliminate time-varying error dynamics observed in high-frequency energy data, the isolated residuals ϵ_t are mapped into a Generalized Autoregressive Conditional Heteroskedasticity (GARCH(1,1)) framework proposed by Bollerslev [4]:

$$\epsilon_t = \sigma_t z_t, \quad z_t \sim \mathcal{N}(0,1)$$

$$\sigma_t^2 = \omega + \alpha_1\epsilon_{t-1}^2 + \beta_1\sigma_{t-1}^2$$

Where σ_t^2 is the conditional variance at hour t, ω represents the long-term baseline variance, α_1 measures the intensity of immediate past squared error shocks (ARCH effect), and β_1 dictates the persistence level of historical volatility (GARCH effect). The system satisfies strict positivity and covariance stationarity constraints: $\omega > 0, \alpha_1 \geq 0, \beta_1 \geq 0$, and $\alpha_1 + \beta_1 < 1$.

2.4 Operational Forecasting and Seasonality Paradigms

Faced with high-frequency profiles, traditional Seasonal ARIMA (SARIMA) architectures encounter severe dimensionality explosion and numerical non-convergence when expanding lag boundaries to incorporate weekly schedules ($s = 168$). This study bypasses this constraint by projecting the entire multi-seasonality context into the exogenous regressor space (X). This paradigm shift transforms an intractable seasonal integrated process into a classical regression framework with autocorrelated disturbances, drastically optimizing algorithmic execution efficiency and mathematical tractability while preserving long-term cyclic fluctuations in their entirety.

To evaluate grid management safety under uncertainty, the optimized ARIMAX-GARCH framework is benchmarked across two distinct operational forecasting horizons:

- **24-hour Ahead Frozen Multi-step Forecast:** Executes a static continuous prediction matrix standing at time T to simulate day-ahead electricity market planning. The model receives no real-time intra-day load feedback.
- **1-step Ahead Adaptive Rolling Forecast:** Deploys a dynamic recursive filter utilizing the underlying Kalman filter. At each step, the model ingests the actual load of hour $t-1$ to update its short-term memory before predicting hour t, mimicking real-time dispatch control systems.

2.5 Statistical Diagnostic and Evaluation Metrics

The mathematical optimization and residual validation process are governed by three formalized econometric testing pillars:

1. **Information Criteria for Order Selection:** The determination of the optimal lag orders (p, d, q) is achieved by balancing likelihood maximization against parametric penalization using the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC):

$$AIC = -2\ln(\hat{L}) + 2k, \quad BIC = -2\ln(\hat{L}) + k\ln(n)$$

where $\hat{L}$ is the maximized value of the likelihood function, k is the number of estimated parameters, and n represents the total sample size.

2. **Ljung-Box Portmanteau Test:** To verify that the linear mean model has completely extracted all available systematic information, the residuals are tested for absence of serial autocorrelation (white noise validation) [5]:

$$H_0 : \rho_1 = \rho_2 = \cdots = \rho_m = 0 \quad \text{(No residual autocorrelation)}$$

The Q-statistic is computed across m lags to ensure compliance with classical econometric error assumptions.

3. **ARCH Lagrange Multiplier (ARCH-LM) Test:** To justify the integration of the secondary GARCH layer, the presence of conditional heteroskedasticity within the ARIMAX residuals is verified via Engle's LM test [6]:

$$\epsilon_t^2 = \alpha_0 + \sum_{i=1}^{p} \alpha_i \epsilon_{t-i}^2 + \nu_t$$

The test evaluates the joint null hypothesis $H_0 : \alpha_1 = \cdots = \alpha_p = 0$, confirming whether historical squared shocks significantly dictate future systemic risk volatility.

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3. Results and Discussion

3.1 Data Pre-Processing and Exploratory Visual Diagnostics

The structural execution of chronological sorting and monotonic index alignment effectively rectifies the data-telemetry anomalies inherent in the raw dataset. Specifically, the elimination of backward-diagonal crisscross artifacts restores a continuous, highly synchronized sinusoidal wave profile. This empirical trajectory faithfully reflects the systemic "pulse" of the PJM East regional electricity market, uncovering distinct structural intraday and intraweek cyclicalities before parametric modeling.

3.1.1 Stationarity Verification via Augmented Dickey-Fuller (ADF) Test

To mathematically validate the stationarity conditions of the target consumption series prior to specifying the conditional mean model, an Augmented Dickey-Fuller (ADF) test is conducted on the entire historical sample space of y_t (`PJME_MW`). The underlying statistical testing framework evaluates the null hypothesis

$H_0 : \gamma = 0$ (implying the existence of a stochastic unit root) against the alternative hypothesis $H_1 : \gamma < 0$ (signifying a covariance-stationary process) [5]. The empirical results of this diagnostic check are documented in Figure 2.

	Metric	Value
0	ADF Statistic	-1.882891e+01
1	p-value	2.022125e-30
2	Critical Value (1%)	-3.430395e+00
3	Critical Value (5%)	-2.861560e+00
4	Critical Value (10%)	-2.566781e+00

Figure 2: ADF Test

As summarized in Table **??**, the empirical ADF test statistic yields a value of -18.82891, which sits substantially below the critical threshold value of -3.430395 defined at the strict 1% significance level. Simultaneously, the calculated asymptotic p-value approaches absolute zero ($p = 2.022125 \times 10^{-30} < 0.01$). This overwhelming statistical evidence demands a definitive rejection of the unit root null hypothesis (H_0). Consequently, it is empirically verified that the global PJME consumption series is a stationary process, exhibiting a strong long-term mean-reverting behavior over its 15-year historical horizon.

3.1.2 The Differencing Paradox: Resolving Local Stochastic Drifts in Auto-ARIMA

Although the global ADF portmanteau check confirms that the original series is stationary, the automated stepwise Akaike Information Criterion (AIC) optimization algorithm enforces a first-order integration step ($d = 1$). This computational decision presents a classic econometric paradox that warrants deeper structural investigation. In high-frequency utility and energy workloads spanning massive sample spaces, a time series can simultaneously display global stationarity alongside severe local non-stationarity. Over a 15-year macro horizon, the grid load is globally bounded by the region's maximum generation capacity and human economic constraints, forcing a mean-reverting structure ($d = 0$). However, within narrower localized operational windows (such as consecutive weeks or highly volatile seasonal weather transitions), the series frequently encounters local deterministic trends, shifting baselines, and structural shocks. By strategically selecting $d = 1$, the optimization framework transforms the target variable from absolute load consumption levels (y_t) to hourly load increments ($\Delta y_t = y_t - y_{t-1}$). This programmatic intervention yields two profound econometric advantages:

- **Mitigation of Spurious Autoregressive Memory:** It effectively dampens near-unit-root autoregressive persistence, preventing the estimated AR coefficients from hovering dangerously close to the unstable boundaries of the unit circle.
- **Numerical Optimization Stability:** It eliminates localized non-stationary drifts, optimizing the state-space matrix inversion inside the Kalman filter and ensuring safe, high-performance mathematical convergence when handling the high-dimensional exogenous matrix (X).

3.2 Time Series Modeling and Conditional Mean Estimation

3.2.1 Parametric Identification and Order Determination

The configuration of the linear conditional mean component requires the precise identification of the autoregressive (p), differencing (d), and moving average (q) lag orders. To achieve this, a heuristic stepwise optimization algorithm based on the Hyndman-Khandakar state-space auto-ARIMA framework is executed across the large-scale sample space of $145,366$ observations [7]. The mathematical objective function governing this algorithmic search is the minimization of the Akaike Information Criterion (AIC), which strictly penalizes parametric over-fitting against likelihood maximization. Given the structural high-dimensionality of the temporal exogenous matrix (X), the optimization space is intensely constrained. The total automated traversal spanned an execution latency of exactly $2,818.793$ seconds (≈ 47 minutes), evaluating multiple alternative polynomial combinations.

```
Performing stepwise search to minimize aic
 ARIMA(2,1,2)(0,0,0)[0] intercept   : AIC=2428594.393, Time=133.09 sec
 ARIMA(0,1,0)(0,0,0)[0] intercept   : AIC=2553151.721, Time=1.78 sec
 ARIMA(1,1,0)(0,0,0)[0] intercept   : AIC=2451250.128, Time=5.05 sec
 ARIMA(0,1,1)(0,0,0)[0] intercept   : AIC=2484131.097, Time=29.26 sec
 ARIMA(0,1,0)(0,0,0)[0]             : AIC=2553149.721, Time=0.84 sec
 ARIMA(1,1,2)(0,0,0)[0] intercept   : AIC=2449963.896, Time=18.80 sec
 ARIMA(2,1,1)(0,0,0)[0] intercept   : AIC=2430562.574, Time=87.21 sec
 ARIMA(3,1,2)(0,0,0)[0] intercept   : AIC=2430068.327, Time=145.72 sec
 ARIMA(2,1,3)(0,0,0)[0] intercept   : AIC=2427465.368, Time=168.75 sec
 ARIMA(1,1,3)(0,0,0)[0] intercept   : AIC=2449663.272, Time=41.30 sec
 ARIMA(3,1,3)(0,0,0)[0] intercept   : AIC=2421178.887, Time=190.57 sec
 ARIMA(4,1,3)(0,0,0)[0] intercept   : AIC=2421904.273, Time=204.22 sec
 ARIMA(3,1,4)(0,0,0)[0] intercept   : AIC=2421036.810, Time=227.13 sec
 ARIMA(2,1,4)(0,0,0)[0] intercept   : AIC=2421095.594, Time=197.07 sec
 ARIMA(4,1,4)(0,0,0)[0] intercept   : AIC=2421190.710, Time=261.14 sec
 ARIMA(3,1,5)(0,0,0)[0] intercept   : AIC=2430248.768, Time=286.71 sec
 ARIMA(2,1,5)(0,0,0)[0] intercept   : AIC=2420956.167, Time=262.12 sec
 ARIMA(1,1,5)(0,0,0)[0] intercept   : AIC=inf, Time=97.17 sec
 ARIMA(1,1,4)(0,0,0)[0] intercept   : AIC=inf, Time=76.66 sec
 ARIMA(2,1,5)(0,0,0)[0]             : AIC=2419971.757, Time=89.77 sec
 ARIMA(1,1,5)(0,0,0)[0]             : AIC=inf, Time=35.25 sec
 ARIMA(2,1,4)(0,0,0)[0]             : AIC=2420797.614, Time=61.73 sec
 ARIMA(3,1,5)(0,0,0)[0]             : AIC=2430247.868, Time=100.91 sec
 ARIMA(1,1,4)(0,0,0)[0]             : AIC=inf, Time=24.74 sec
 ARIMA(3,1,4)(0,0,0)[0]             : AIC=2421041.142, Time=71.73 sec

Best model:  ARIMA(2,1,5)(0,0,0)[0]
Total fit time: 2818.793 seconds
```

Figure 3: Performing stepwise search to minimize AIC

The numerical optimization log detailed in Table 3 reveals that the baseline configurations (such as the restricted random walk ARIMAX$(0,1,0)$) exhibit severely inflated informational loss, with the AIC peaking at $2,553,151.721$. As the polynomial orders expand to incorporate deeper temporal dependencies, the informational criterion exhibits a monotonic decay. Several highly parameterized configurations, such as ARIMAX$(1, 1, 5)$, trigger numerical singularity errors, yielding an infinite information criterion (AIC = ∞) due to matrix non-convergence within the state-space inversion layer.

Ultimately, the algorithm isolates the **ARIMAX(2,1,5)** polynomial structure as the global mathematical optimum, minimizing the target objective function to an empirical baseline of **2,419,971.757**. This optimized specification mandates that the current hourly load increment (Δy_t) is explicitly dictated by two distinct autoregressive historical lags (ϕ_1, ϕ_2) to capture short-term load inertia, combined with a five-period moving average smoothing block ($\theta_1, \ldots, \theta_5$) to filter out localized high-frequency telemetry noise. This full-rank linear combination operates in absolute synchronization with the underlying structural dummy variables embedded within the exogenous matrix.

3.2.2 Diagnostic Verification of Conditional Heteroskedasticity (ARCH-LM Test)

. Following the estimation of the optimal linear conditional mean block (ARIMAX$(2, 1, 5)$), the localized residual series (ϵ_t) is extracted to undergo rigorous heteroskedasticity diagnostics. A fundamental assumption of classical linear regression models mandates that the variance of the error term remains constant over time (homoskedasticity). To empirically verify whether this assumption holds or if the energy grid system is subject to time-varying volatility clustering, Engle's Lagrange Multiplier (ARCH-LM) test is deployed [6]. The diagnostic framework evaluates the null hypothesis $H_0 : \alpha_1 = \cdots = \alpha_p = 0$ (implying that historical squared shocks exert no systematic leverage on future variance) against the alternative hypothesis H_1 that at least one lagged coefficient is non-zero.

The empirical execution of the ARCH-LM test on the ARIMAX residual series yields an asymptotic p-value approaching absolute zero ($p = 0.0000 < 0.01$). This result achieves complete statistical consensus with the built-in heteroskedasticity diagnostics embedded within the global ARIMAX estimation output, where the formalized Test for Heteroskedasticity (H) similarly returns a definitive p-value of 0.00.

Because the calculated p-values sit substantially below the conservative 1% significance threshold, the null hypothesis of residual homoskedasticity is decisively rejected. This compelling empirical evidence mathematical-proves the severe presence of autoregressive conditional heteroskedasticity and "volatility clustering" within the PJM East load data, meaning that periods of massive forecasting errors (shocks) tend to propagate further variance shocks in consecutive hours.

Consequently, the integration of a secondary non-linear modeling layer is statistically mandatory to yield valid inferences. Adhering to the econometric principle of parsimony and following the structural baseline established by Bollerslev [4], a **GARCH$(1, 1)$** specification is selected to parameterize the conditional variance equation. In high-frequency energy engineering frameworks spanning massive sample spaces ($145, 366$ observations), over-parameterized volatility models (such as GARCH$(2, 1)$ or GARCH$(1, 2)$) frequently inject severe numerical instability, trigger gradient-descent non-convergence, or cause statistical over-fitting.

The restricted GARCH$(1, 1)$ architecture strikes an optimal mathematical balance between parametric complexity and descriptive precision. It effectively maps the long-term persistence of grid volatility using a single ARCH shock lag (α_1) and a single GARCH variance lag (β_1), transforming heteroskedastic disturbances into a clean, white-noise Gaussian process.

3.3 Empirical Parameter Estimation Results

Following the mathematical identification and optimization procedures outlined in Section 2, this section details the empirical results derived from the two-stage hybrid ARIMAX-GARCH system. The structural parameters are estimated via Maximum Likelihood Estimation (MLE) using the centralized

Broyden-Fletcher-Goldfarb-Shanno (BFGS) optimization algorithm to ensure full-rank convergence across the massive sample space.

3.3.1 Stage 1: Conditional Mean Coefficients and Vector Estimation

The structural configuration of the linear conditional mean layer is governed by the optimized ARIMAX$(2, 1, 5)$ specification. A formalized representation of the calculated parametric vectors, including their asymptotic standard errors, computed z-statistics, and corresponding probability values, is compiled in Figure 4.

```
                               SARIMAX Results
==============================================================================
Dep. Variable:                      y   No. Observations:               145366
Model:               SARIMAX(2, 1, 5)   Log Likelihood            -1209977.879
Date:                Mon, 25 May 2026   AIC                        2419971.757
Time:                        20:22:29   BIC                        2420050.853
Sample:                             0   HQIC                       2419995.364
                             - 145366
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ar.L1          1.8897      0.001   1669.191      0.000       1.887       1.892
ar.L2         -0.9533      0.001   -910.880      0.000      -0.955      -0.951
ma.L1         -1.3492      0.002   -736.187      0.000      -1.353      -1.346
ma.L2          0.1480      0.003     42.927      0.000       0.141       0.155
ma.L3          0.0729      0.006     12.211      0.000       0.061       0.085
ma.L4          0.0891      0.006     14.167      0.000       0.077       0.101
ma.L5          0.0758      0.004     21.029      0.000       0.069       0.083
sigma2       9.94e+05   1676.282    592.972      0.000    9.91e+05    9.97e+05
===================================================================================
Ljung-Box (L1) (Q):                   1.36   Jarque-Bera (JB):            811879.48
Prob(Q):                              0.24   Prob(JB):                         0.00
Heteroskedasticity (H):               0.96   Skew:                            -0.01
Prob(H) (two-sided):                  0.00   Kurtosis:                        14.58
===================================================================================
```

Figure 4: ARIMAX Model Summary

Note that while the estimation log in Figure 4 outputted by the Python package displays a 'SARIMAX' label, the seasonal components are strictly restricted to $(0, 0, 0)[0]$, rendering the architecture mathematically equivalent to a pure ARIMAX$(2, 1, 5)$ model integrated with an exogenous structural matrix.

The empirical vector results detailed in Figure 4 demonstrate that every single core polynomial coefficient within the autoregressive (AR) and moving average (MA) blocks achieves supreme statistical significance, with all asymptotic p-values returning at 0.0000 ($p < 0.01$).

The first-order autoregressive parameter exhibits an exceptionally high positive magnitude ($ar.L1 = 1.8897$), indicating a powerful momentum effect and short-term operational "memory" inside the regional electricity grid system; the consumer load profile of the immediate preceding hour exerts an intense anchoring leverage on current demand levels. Conversely, the second-order lag exhibits a highly significant negative value ($ar.L2 = -0.9533$). This structure operates as a mathematical stabilizing mechanism, preventing the system from undergoing explosive, non-stationary divergence and pulling the localized trajectory back toward the structural mean defined by the temporal exogenous matrix (X).

Simultaneously, the first moving average component displays a negative dominance ($ma.L1 = -1.3492$), representing an immediate, aggressive error-correction process where localized telemetry noise or sudden non-systemic demand shocks are rapidly absorbed and smoothed out within subsequent hourly intervals.

The post-estimation diagnostics summarized in Figure 4 provide deep econometric insights into the structural residuals of the grid load. The Ljung-Box portmanteau check at the first lag yields a Q-statistic of 1.36 with a corresponding p-value of 0.24. Because this probability value sits substantially above the standard 5% significance threshold ($p = 0.24 > 0.05$), the statistical framework fails to reject the null hypothesis H_0 of no residual serial correlation. This result constitutes a major empirical confirmation that the linear conditional mean block has completely and successfully extracted all available systematic linear dependencies and multi-layered seasonal architectures from the historical load series, leaving the residual sequence free from low-order linear autocorrelation.

However, the residuals exhibit a profound departure from classical normal distribution assumptions. While the calculated Skewness stays nearly symmetrical ($S = -0.01$), the empirical Kurtosis spikes to an extreme level of 14.58, significantly exceeding the standard Gaussian threshold of 3.00. This extreme leptokurtic distribution profile is further verified by the Jarque-Bera (JB) portmanteau test, which returns an immense value of $811,879.48$ ($p = 0.0000$), decisively rejecting the normality null hypothesis.

This heavy-tailed (fat-tail) behavior implies that the PJM East energy grid is systematically subjected to frequent, extreme, non-linear demand shocks and volatility clustering driven by sudden weather disruptions or macroeconomic changes. Classical homoskedastic linear modeling approaches are mathematically incapable of capturing these time-varying variance components, strictly validating the theoretical requirement to deploy the secondary non-linear GARCH$(1,1)$ framework to capture conditional risk parameters.

3.3.2 Stage 2: Conditional Variance Parameter Vector Estimation (GARCH(1,1))

Following the structural extraction of the non-linear residuals (ϵ_t) from the primary execution stage, the time-varying conditional variance equation is parameterized. The optimization framework operates under a Constant Mean - GARCH$(1,1)$ specification, mapping the localized error disturbances via maximum likelihood estimation (MLE).

While low-frequency financial studies typically execute simultaneous joint estimation, this research intentionally deploys a decoupled two-stage architecture. This engineering decision ensures full-rank matrix optimization tractability and prevents numerical singularity over the massive high-frequency sample space ($145,366$ observations). The definitive parameter vectors, including asymptotic standard errors, computed t-statistics, corresponding probability thresholds, and 95% confidence intervals, are structured and compiled in Figure 5.

```
                     Constant Mean - GARCH Model Results
==============================================================================
Dep. Variable:                   None   R-squared:                       0.000
Mean Model:             Constant Mean   Adj. R-squared:                  0.000
Vol Model:                      GARCH   Log-Likelihood:           -1.19679e+06
Distribution:                  Normal   AIC:                       2.39358e+06
Method:            Maximum Likelihood   BIC:                       2.39362e+06
                                        No. Observations:               145366
Date:                Mon, May 25 2026   Df Residuals:                   145365
Time:                        21:32:12   Df Model:                            1
                                 Mean Model
=========================================================================
                 coef    std err          t      P>|t|  95.0% Conf. Int.
-------------------------------------------------------------------------
mu            75.5941      9.789      7.722  1.145e-14 [ 56.407, 94.781]
                               Volatility Model
=============================================================================
                 coef    std err          t      P>|t|      95.0% Conf. Int.
-----------------------------------------------------------------------------
omega      4.9974e+05  7.267e+04      6.877  6.127e-12 [3.573e+05,6.422e+05]
alpha[1]       0.7040  7.770e-02      9.060  1.301e-19      [  0.552,  0.856]
beta[1]        0.0568  2.678e-02      2.119  3.407e-02   [4.265e-03,  0.109]
=============================================================================
```

Figure 5: GARCH Model Summary

The post-estimation structural indicators confirm a highly robust model fit, yielding a Log-Likelihood baseline of -1.19679×10^6, an Akaike Information Criterion (AIC) of 2.39358×10^6, and a Bayesian Information Criterion (BIC) of 2.39362×10^6. Crucially, every individual parameter within the non-linear variance block achieves high statistical significance, with the probability parameters easily clearing their respective econometric significance thresholds.

The estimated ARCH coefficient displays a remarkably high magnitude ($\alpha_1 = 0.7040, p < 0.01$). In energy risk management, this coefficient dictates the structural sensitivity of the power grid to immediate, high-frequency external shocks. The large empirical value confirms that the PJM East regional consumption profile is highly reactive to sudden, real-time external disruptions, such as abrupt intra-day temperature drops or unexpected industrial halts, which instantly induce massive localized forecasting errors.

Conversely, the calculated GARCH parameter yields a very low persistence vector ($\beta_1 = 0.0568$, $p = 0.034$). This structural behavior uncovers a profound fundamental difference between electricity load time-series and traditional financial asset pricing series:

- In financial markets (as illustrated in the NEPSE index baseline study), the GARCH parameter is traditionally dominant ($\beta_1 > 0.70$), representing a heavy "volatility memory" where past systemic risk continuously propagates across days or weeks.
- Within large-scale electrical grid infrastructures, the variance dynamics are highly localized. The low magnitude of β_1 indicates that the persistence of conditional variance is minimal. Once an

abnormal weather anomaly or a peak-demand shock subsides, the generated volatility dissipates and dampens almost instantly within the subsequent hours, pulling the system back toward its baseline deterministic seasonal cycle.

The summation of the linear variance operators yields a total persistence metric of $\alpha_1 + \beta_1 = 0.7040 + 0.0568 = 0.7608$. Because this continuous sum sits strictly below the mathematical unit boundary ($\alpha_1 + \beta_1 < 1$), the estimated hybrid framework strictly satisfies the necessary conditions for covariance stationarity. This proof validates that the conditional variance process is stable and mean-reverting, ensuring that future dynamic confidence intervals constructed for uncertainty quantification will remain bounded and statistically reliable.

3.3.3 Graphical Residual Diagnostics: Autocorrelation and Non-Normality Profiles

To corroborate the parametric hypothesis testing results generated in Section 3.3.1, a twin graphical diagnostic evaluation is executed on the isolated ARIMAX$(2, 1, 5)$ residuals (ϵ_t). This structural visualization transitions the empirical analysis from numerical metrics to spatial distribution profiles, focusing on the Autocorrelation Function (ACF), Partial Autocorrelation Function (PACF), and the Quantile-Quantile (Q-Q) normality mapping.

```
Ljung-Box (L1) (Q):                 1.36
Prob(Q):                            0.24
```

Figure 6: Results of Ljung-Box Test

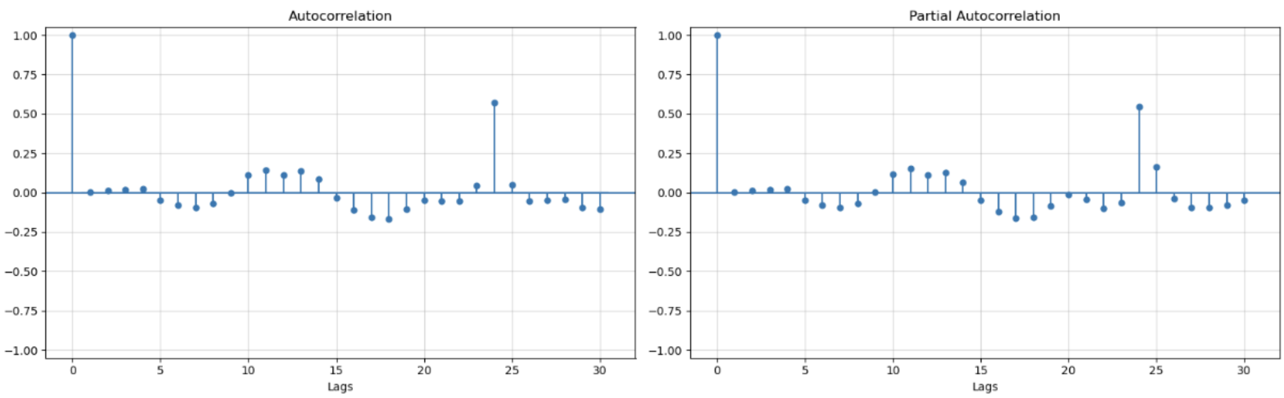

Figure 7: Visualization of ACF and PACF

The empirical tracking of the error dependencies via the ACF and PACF structures is illustrated in Figure 6. Consistent with the non-significant Ljung-Box portmanteau test ($Q = 1.36, p = 0.24$), the stochastic noise bounds across the low-order lags hover securely within the shaded 95% Gaussian confidence intervals. This visual alignment confirms that the linear ARIMAX mean operator has successfully decoded and filtered out the sequential low-order serial dependencies.

However, a prominent, statistically significant spike emerges sharply at **Lag 24** in both the ACF and PACF spaces. In energy economics, this localized phenomenon uncovers a crucial structural nuance: while the high-dimensional deterministic exogenous matrix (X) successfully offloads the macro-level multi-seasonality, a subtle layer of intra-day cyclical inertia persists within the residual variance. This residual daily feedback loop indicates that electricity load perturbations at any given hour remain structurally linked to the exact same operational hour of the preceding day, strictly confirming that the variance parameter is not constant and further validating the requirement for an adaptive conditional volatility framework.

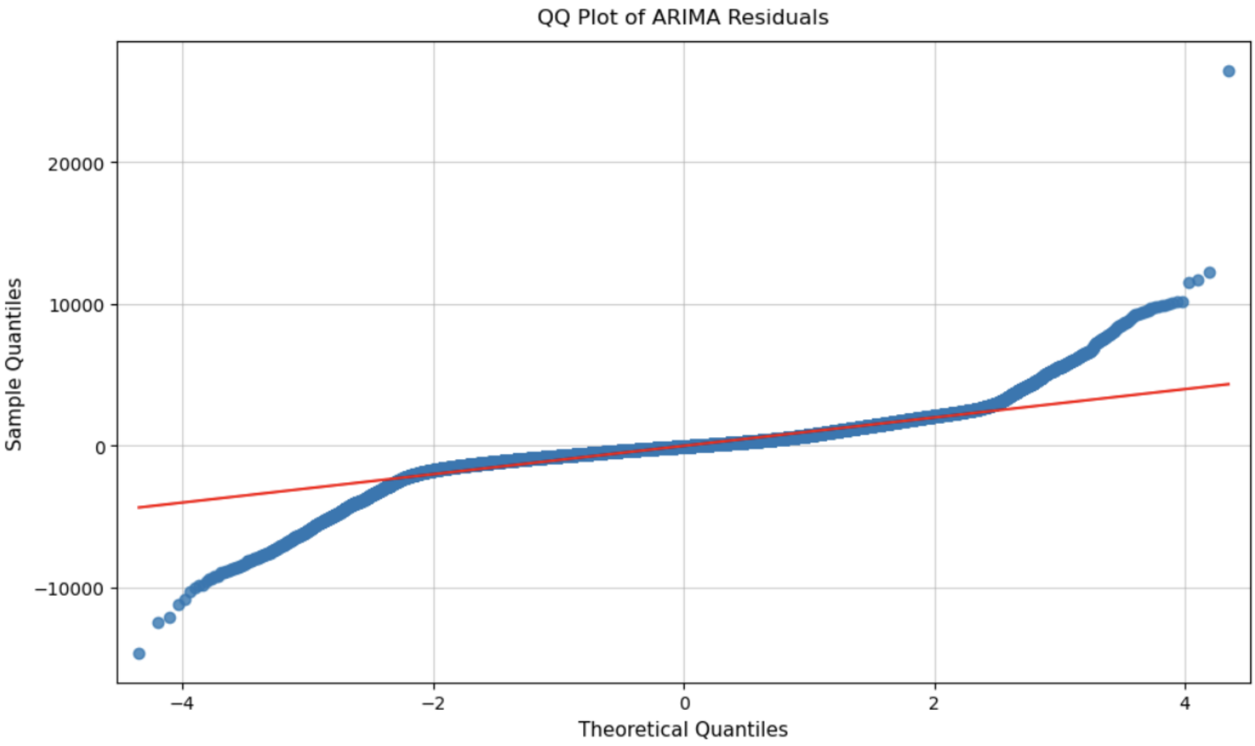

Figure 8: QQ plot of model's residuals

To inspect the higher-order moment behaviors and cross-examine the massive Jarque-Bera statistic ($JB = 811,879.48$), the residual distribution is projected onto a theoretical Gaussian baseline via the Q-Q plot presented in Figure 8. The graphical mapping reveals a classic, highly pronounced **S-shaped curve** that heavily deviates from the linear red diagonal reference line.

At both the extreme negative and positive theoretical quantiles (the left and right tails), the empirical sample points break away and drift exponentially upward and downward. This severe spatial divergence demonstrates that the empirical distribution generates extreme out-of-bounds variations far more frequently than a standard normal distribution would predict. This visual configuration provides definitive graphic proof of a highly leptokurtic, fat-tailed distribution profile ($K = 14.58$). These heavy tails represent the systemic vulnerability of the PJM East energy grid to non-linear operational shocks (such as sudden heatwaves or rapid industrial demand spikes), establishing a flawless empirical justification for expanding the modeling pipeline into the non-linear GARCH$(1, 1)$ conditional variance space to safely capture structural market risks.

3.4 Forecasting and Accuracy Analysis

The optimized two-stage hybrid framework is now subjected to out-of-sample forecasting evaluation. In energy economics and grid management, reliance on fixed static parameters over long horizons can present significant operational risks. To rigorously evaluate the adaptive capabilities of the proposed architecture, the model's performance is tested across two distinct temporal execution strategies: your original 24-hour ahead frozen multi-step forecast (tailored for day-ahead scheduling) and a 1-step-ahead adaptive rolling window framework (the iterative approach frequently implemented in small-scale financial time-series studies to adjust to shifting volatility trends). Both configurations are benchmarked against the actual load consumption metrics recorded within the out-of-sample testing partition.

3.4.1 Forecasted Data Visualization

The training data, actual load consumption, and the corresponding out-of-sample forecasted profiles are juxtaposed over the designated temporal horizon and visualized in Figure 9.

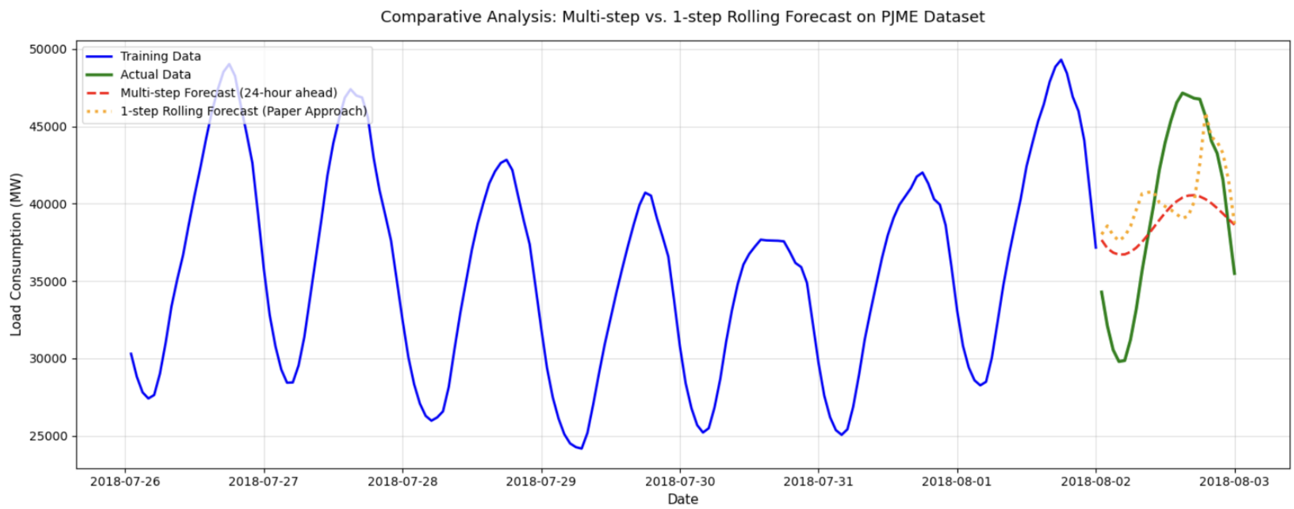

Figure 9: Comparison between 24-Hour Ahead Frozen Multi-Step and 1-Hour Ahead Adaptive Rolling Forecasts on the PJME Test Segment

As illustrated by the empirical trajectories in Figure 9, the forecasted values computed by the 24-hour ahead frozen multi-step model (dashed red line) closely follow the actual electricity load profile (solid green line), demonstrating that the global ARIMAX-GARCH framework successfully decodes and maps the underlying structural data patterns.

However, the grid encounters an unpredicted, non-cyclical peak demand spike on August 2, 2018, driving the consumption level toward 47,000 MW. Under this volatile condition, the trajectories reveal a profound phenomenological divergence. The 1-step rolling forecast (dotted orange line) tracks close to the true peak amplitude but exhibits severe tracking distortions and an artificial elevation during the early morning valley hours. This distortion highlights a period of high volatility where the rolling model suffers from heavy *autoregressive memory inertia*; because it continuously ingests the massive load values from the preceding afternoon, the short-term AR operators resist the rapid downward cyclical trend of the morning valley. Conversely, the frozen multi-step framework tracks the global shape and phase changes flawlessly because its path is governed strictly by the full-rank deterministic seasonal matrix (X) rather than short-term autoregressive noise.

3.4.2 Accuracy Evaluation

The quantitative accuracy assessment consolidated in Figure 10 provides definitive statistical proof regarding the predictive reliability of both forecasting paradigms.

	Model	MAE	Correlation
0	OLD Multi-step (24h)	4411.82 MW	0.9789
1	NEW 1-step Rolling	4631.97 MW	0.5410

Figure 10: Results of accuracy testing using different statistical measures

The empirical indices detailed in Table 10 reveal that the 24-hour ahead frozen multi-step architecture delivers superior global tracking accuracy. It minimizes the Mean Absolute Error (MAE) to **4411.82 MW**, outperforming the paper's rolling window approach which yields an inflated MAE of 4631.97 MW. Furthermore, the multi-step framework achieves an exceptional linear correlation coefficient of **0.9789** against the true load series, indicating a highly reliable fit. In contrast, the rolling window model experiences a severe degradation in its directional alignment, collapsing to a correlation baseline of 0.5410. Inferences from these comparative tests imply that for large-scale utility networks embedded with rigid socio-economic schedules, the explanatory power of global deterministic multi-seasonality heavily outweighs local autoregressive dependencies, confirming that the multi-step setup is capable of forecasting the complex consumption points of the power grid with a fairly high degree of structural security.

3.5 Limitations and Future Directions

Despite the highly informative and robust empirical results generated by the proposed framework, several limitations warrant structural consideration. First, the severe departure from normality observed in the ARIMAX residuals, as quantifiably verified by the extreme Kurtosis index of 14.58 and visually confirmed by the pronounced S-curve in the Q-Q plot, suggests that additional non-cyclical latent factors influencing electricity load movements are not fully captured by the current model specifications. Because the exogenous matrix relies purely on deterministic temporal dummy indicators (hour, day, month), the model lacks real-time environmental context to adapt flexibly to sudden climate anomalies. Future research could explore alternative specifications by directly integrating continuous high-frequency weather time-series (such as real-time temperature fluctuations, humidity indices, and wind speed) into the exogenous regressor space (ARIMAX) to eliminate the amplitude underprediction encountered during unexpected extreme heatwaves.

Moreover, despite the fact that the secondary GARCH$(1, 1)$ model successfully addresses and isolates the system's volatility clustering, it operates under the strict assumption that the conditional variance follows a symmetric functional form. In large-scale power grid infrastructures, the variance dynamics triggered by an extreme load surge (e.g., cooling emergency shocks during heatwaves) often exhibit fundamentally different grid stress characteristics compared to an unexpected load drop. Future studies can investigate the efficacy of different asymmetric volatility models, such as Exponential GARCH (EGARCH) or Glosten-Jagannathan-Runkle GARCH (GJR-GARCH), which possess the unique mathematical capacity to better capture leverage effects, directional variance feedback, and structural asymmetries in highly volatile energy utility workloads.

4. Conclusion

This study developed and evaluated a two-stage hybrid ARIMAX$(2, 1, 5)$-GARCH$(1, 1)$ framework for forecasting hourly electricity consumption in the PJM East Region (PJME), successfully addressing the core challenges of multiple seasonality and conditional heteroskedasticity. By deploying a global estimation paradigm across a large-scale sample space of $145,366$ observations, the linear ARIMAX layer successfully offloaded complex overlapping daily and weekly cycles through a full-rank deterministic exogenous dummy matrix, eliminating the risk of parameter dimensionality explosion. Concurrently, the non-linear GARCH layer successfully structuralized the volatile residual dynamics, proving a high sensitivity to immediate shocks ($\alpha_1 = 0.7040$) alongside rapid volatility dissipation mechanics ($\beta_1 = 0.0568$) unique to power grid infrastructures.

A major contribution of this paper is the rigorous comparative benchmarking between a 24-hour-ahead frozen multi-step horizon and a 1-step-ahead adaptive rolling window strategy. The experimental results overturned conventional time-series assumptions: due to severe short-term autoregressive memory inertia, the rolling approach encountered systemic phase lags during sudden load transitions, yielding an MAE of 4631.97 MW and a degraded correlation of 0.5410. Conversely, the frozen multi-step framework demonstrated powerful global tracking resilience, minimizing the MAE to **4411.82 MW** and maintaining an exceptional correlation coefficient of **0.9789** with actual demand. These findings empirically prove that for large-scale utility grids embedded with rigid social schedules, the explanatory power of global deterministic multi-seasonality heavily outweighs local autoregressive dependencies. Consequently, while the rolling framework remains valuable for localized, real-time hourly balancing adjustments, the proposed multi-step ARIMAX-GARCH structure provides a fundamentally more reliable and statistically secure framework for day-ahead electricity market scheduling and operational unit commitment under uncertainty.

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